PhyST-Net: A Physics-Aware Differentiable Spatio-Temporal Transformer for Constrained Network Forecasting

Xu Li

***Abstract*—Spatio-temporal forecasting is a core predictive task for complex graph-structured systems, which is important for operational decision-making in domains such as intelligent traffic routing and resource scheduling. However, existing methodologies frequently blur the temporal distinction between historical observations and future exogenous drivers, while struggling to natively satisfy strict physical operational constraints without relying on naive post-hoc clipping that may weaken gradient signals or create train-inference mismatch. To overcome these limitations, we propose PhyST-Net, a generalized exogenous-aware spatio-temporal graph Transformer framework for domain-constrained network forecasting. Specifically, the framework introduces Adaptive Gaussian-soft-windowed Patching (AGSP) for continuous temporal tokenization and a Relational Spatio-Temporal Manifold Fusion (R-STMF) adapter that utilizes a global token to efficiently decouple macro-trend and local-variation spatial dependencies. Furthermore, it incorporates KAN-enhanced Adaptive Modulation and Conditioning (K-AMC) for non-linear modulation of future exogenous factors, and a Physics-Informed Dynamic Capping Head (PI-DCH) that employs a differentiable barrier gate to promote constraint-aware decoding. The proposed formulation provides a reusable basis for evaluating domain-constrained forecasting systems under diverse operational regimes.**

***Index Terms*—Spatio-temporal forecasting, graph Transformers, exogenous variables, domain-constrained machine learning, network forecasting.**

I. INTRODUCTION

SPATIO-TEMPORAL forecasting predicts future states in graph-structured systems (e.g., transportation networks, power grids) where temporal dynamics and spatial interactions evolve jointly. Node-level trajectories are rarely self-driven; rather, they are shaped by historical conditions, future external drivers, and domain-specific feasibility constraints. As illustrated in Fig. 1, complex network signals comprise shared macro-trends driven by exogenous forcing and high-frequency local variations. Consequently, a practical forecasting model must learn continuous temporal representations, infer dynamic graph relations, and integrate future exogenous variables (ExG) without data leakage, all while maintaining operational compatibility.

A. The Forecasting Trilemma in Spatio-Temporal Networks

Despite recent architectural advancements, modern spatio-temporal forecasting frameworks remain hampered by a "Forecasting Trilemma"—an inherent trade-off between Model Capacity, Domain Consistency, and Spatio-Temporal Complexity.

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1) Model Capacity vs. High-Frequency Noise: Capacity denotes architectural expressive power. While modern Transformers excel at sequence modeling, they are highly sensitive to high-frequency stochastic noise in raw spatio-temporal data. Furthermore, rigid temporal patching exacerbates this by artificially splitting continuous physical regimes. As highlighted by recent trends in complexity-balanced sequence modeling, uniform representational capacity across an entire time horizon is computationally inefficient. The challenge is designing an architecture capable of capturing macro-trends while employing adaptive temporal allocation to filter node-specific noise.

2) Domain Consistency and Constraint-Aware Operations: Consistency mandates that a model's outputs adhere to the fundamental physical laws or operational limits inherent to the target domain. Purely data-driven architectures frequently generate unfeasible predictions—such as negative vehicular speeds or resource usage exceeding an operational upper bound. Such violations severely degrade operator trust in critical infrastructure. Drawing from emerging paradigms in safety-critical generation that utilize output-space guidance, a robust forecasting framework must enforce physical feasibility directly within its decoding pathway, moving beyond the limitations of reactive, post-hoc corrections.

3) Spatio-Temporal Complexity and Multi-Scale Dynamics: Complexity involves the management of dense spatial couplings across heterogeneous nodes. In physical networks, nodes interact via multiple simultaneous mechanisms: local topological dependencies (e.g., localized propagation effects, adjacent road segments) and global environmental fronts (e.g., system-wide contextual regimes). Existing models frequently conflate these multi-scale interactions into a singular spatial operator, thereby failing to distinguish localized perturbations from system-wide drivers.

To systematically resolve this trilemma, we propose PhyST-Net, an ExG-aware Spatio-Temporal Graph Transformer designed as a versatile framework for domain-constrained network forecasting. To accommodate the multi-scale and multi-modal nature of dynamic network signals, we introduce the following key innovations:

- **Continuous and Differentiable Patching**: We present the **Adaptive Gaussian-soft-windowed Patching (AGSP)** mechanism. By transforming discrete observations into learnable spectral-temporal densities, AGSP recovers continuous event dynamics and functions as an adaptive low-pass filter, effectively rejecting high-frequency noise.
- **Decoupled Spatio-Temporal Topology**: We introduce the **Relational Spatio-Temporal Manifold Fusion (R-STMF)** framework. R-STMF leverages a dual-branch

graph logic to decompose latent node dynamics into multi-scale spatial manifolds, mitigating the spatial conflation typical of standard distance-based graphs.
- **Piecewise Conditioning and Safe Capping**: We implement the **KAN-enhanced Adaptive Modulation and Conditioning (K-AMC)** module, utilizing Kolmogorov-Arnold spline edge functions for the non-linear modulation of graph-enhanced states via exogenous drivers. This is coupled with a **Physics-Informed Dynamic Capping Head (PI-DCH)** that embeds strict physical safety constraints and adherence to operational envelopes directly into the output space.

By explicitly partitioning the data flow into distinct computational pathways—historical context, relational reasoning, and exogenous conditioning—PhyST-Net establishes a rigorous forecasting contract. This interface-centric design prevents the entanglement of historical data and future drivers, ensuring the architecture can be seamlessly instantiated across diverse critical domains.

B. Contributions

The primary contributions of this work are summarized as follows:

- **A general ExG-aware graph-forecasting formulation.** We define network forecasting as a multi-modal task where historical states, exogenous variables, and domain constraints are processed through distinguishable pathways, thereby supporting broad architectural generalizability.
- **A continuous temporal tokenization module.** We introduce AGSP, which supersedes rigid patch boundaries with learnable Gaussian soft-windows, preserving event continuity and minimizing boundary artifacts in non-stationary sequences.
- **A modular graph adapter for mixed spatial mechanisms.** We develop R-STMF to disentangle multi-scale spatial dependencies prior to relational propagation, providing a modular contract that accommodates alternative graph predictors.
- **A future-ExG-conditioned decoding path.** We fuse K-AMC and PI-DCH to inject forecast-horizon drivers and structurally promote operational feasibility within the model, bridging latent representation learning with strict operational requirements.

The experimental section later instantiates this formulation on real-world forecasting tasks under diverse operational regimes.

II. Related Work

This section reviews representative research trajectories pertinent to PhyST-Net. Rather than chronicling models chronologically, we categorize prior literature based on the core modeling challenges inherent to ExG-aware spatio-temporal forecasting. A summary of the architectural taxonomies is provided in Table I.

A. Temporal Forecasting and Transformer Architectures

Classical statistical forecasting techniques, including autoregressive and state-space models, retain utility when predicting low-dimensional, approximately stationary processes. However, these foundational assumptions become restrictively fragile in graph-structured systems where node dynamics are highly non-linear and persistently coupled with external forcing. While deep recurrent and convolutional architectures improved non-linear sequence modeling via latent state tracking and local temporal filtering, long-horizon forecasting remains challenging due to error accumulation and vanishing gradients over extended time steps.

Transformer-based forecasters resolve long-range dependency constraints via self-attention, establishing themselves as the dominant paradigm for deep time-series analysis. Early breakthrough models, such as Informer [2], Autoformer [3], and FEDformer [4], focused on reducing the quadratic complexity of attention and decomposing trend-seasonal patterns. Recent variants, utilizing PatchTST-style patching and inverted-dimension attention, significantly reduce computational overhead while enhancing representation learning for multivariate sequences [1], [5], [6]. Nevertheless, temporal patching is predominantly implemented via rigid, fixed-length windows. In highly non-stationary sequences, physical regime transitions and rapid fluctuations rarely align with these artificial boundaries. PhyST-Net advances this trajectory by introducing AGSP, rendering temporal tokenization fully learnable and mathematically continuous.

B. Spatio-Temporal Graph Forecasting

Spatio-temporal graph neural networks (STGNNs) build upon foundational graph convolutions [7] and graph attention mechanisms [8] to capture node interactions. These methodologies are highly effective when cross-node dependencies dictate system behavior, such as in traffic corridors and power grids [9]–[11]. Furthermore, adaptive graph learning relaxes the restrictive assumption of a static, fully observable topology, allowing models to infer latent relations directly from data and dynamically adjust propagation patterns.

A critical limitation, however, is the widespread reliance on homogeneous propagation protocols across all latent states. In physical systems, spatial dependence is inherently multimodal; nodes interact through localized proximities, shared environmental influences, and overarching exogenous drivers. A singular adjacency matrix, even if learned dynamically, conflates these distinct mechanisms. R-STMF resolves this by decomposing temporal representations into high-frequency local variations and low-frequency macro-trends, applying targeted, sparse relational propagation to each manifold prior to fusion.

C. Exogenous-Variable-Aware Forecasting

Real-world forecasting is intrinsically dependent on exogenous (ExG) variables. In vertical domains, such as intelligent traffic routing—where models like ConFormer [12] leverage

TABLE I
TAXONOMY AND COMPARATIVE ANALYSIS OF SPATIO-TEMPORAL FORECASTING MODELS.

Category	Model	Paradigm	Temporal Tokenization	Spatial Interaction	Exogenous Integration	Constraint Awareness
General Transformer Forecasting	Informer [2]	Transformer	Point-wise / ProbSparse	–	Feature concat.	–
	Autoformer [3]	Transformer	Series decomposition	–	Feature concat.	–
	PatchTST [5]	Transformer	Rigid patches	–	–	–
	iTransformer [6]	Transformer	Inverted dimension	Channel mixing	Covariate tokens	–
Spatio-temporal Graph Forecasting	STGCN [9]	ST-GNN	Gated temporal conv.	Static graph conv.	Feature concat.	–
	Graph WaveNet [10]	ST-GNN	Dilated causal conv.	Adaptive graph conv.	Feature concat.	–
	SDG-Net [11]	ST-GNN	Convolutional	Latent dual-graph	–	–
Exogenous-aware Spatio-temporal Forecasting	TimeXer [14]	Hybrid	Rigid patches	Global token / cross-attn.	Variate tokens / cross-attn.	–
	ConFormer [12]	Hybrid	Transformer encoder	Cross-attention	External event emb.	–
	DAG [15]	ST-GNN	Dependency modeling	Directed adaptive graph	ExG-guided dependency	–
	GCGNet [16]	ST-GNN	Generative / recurrent	Consistent alignment	Joint temporal-channel	–
	ExoST [13]	ST-GNN	Convolutional	ExG-driven graph	Select-then-balance	–
Physics-aware Forecasting	GeatFormer [24]	Physics-inf.	Rigid patches	Dual-branch graph	–	Penalty loss
	PGNN-TFT [27]	Physics-inf.	Transformer / TFT	Static domain graph	–	Post-hoc clipping
Ours	**PhyST-Net**	**Hybrid**	**AGSP**	**R-STMF**	**K-AMC**	**PI-DCH**

Notes: "–" indicates that the corresponding mechanism is not explicitly considered. ExG denotes exogenous variables; concat., emb., conv., and attn. denote concatenation, embedding, convolution, and attention, respectively. AGSP, R-STMF, K-AMC, and PI-DCH denote adaptive Gaussian-soft-windowed patching, relational spatio-temporal manifold fusion, KAN-based adaptive modulation conditioning, and physics-informed differentiable constraint head, respectively.

external factors like accidents and regulations—and resource-constrained operational forecasting, ExG features are critical for predictive resilience.

Recent foundational architectures have begun systematically targeting ExG fusion to address inconsistent covariate effects [13]. In time-series modeling, TimeXer [14] empowers Transformers by mapping external factors into variate tokens, enabling cross-attention between endogenous patches and exogenous drivers. Concurrently, DAG [15] introduces dual correlation networks to capture channel-wise dependencies, while in the spatio-temporal graph domain, models like GCGNet [16] employ generative frameworks to align predicted correlations with joint temporal-channel graphs driven by external factors.

While these mechanisms significantly improve accuracy under ideal conditions, they frequently blur the temporal distinction between historical observations and future drivers. Specifically, historical ExG elucidates recent state trajectories, whereas future ExG provides horizon-level modulation. Merging both into a singular feature tensor exacerbates temporal imbalance effects [13] and complicates data availability reasoning at inference time. Consequently, PhyST-Net structurally decouples historical ExG from future ExG. K-AMC leverages the future-ExG pathway to generate latent modulation parameters, enabling forecast-horizon drivers to dynamically condition graph-enhanced states without being erroneously treated as historical priors.

D. Interpretable Conditional Modeling with KANs

Interpretability is paramount in operational forecasting, as dispatchers require insight into the primary drivers shaping a prediction. While attention weights and post-hoc attribution methods offer diagnostic utility, they often fail to reflect the model's actual computational pathways. Kolmogorov-Arnold Networks (KAN) [19] represent a paradigm shift from traditional perceptrons by modeling transformations via learnable, univariate spline-based edge functions [20]–[23].

By structuring a conditioning network as an additive sum of feature-wise edge functions, the influence of individual input variables can be analytically inspected through learned basis coefficients and derivatives. PhyST-Net capitalizes on this property within the K-AMC module: future ExG features are mapped to scale and shift parameters via spline functions. While not establishing strict causal identifiability, this architectural alignment provides a transparent, inspectable explanation mechanism far superior to generic dense concatenation.

E. Physics-Informed and Constraint-Aware Forecasting

Physics-informed learning embeds essential domain knowledge into neural architectures via structural priors, penalty losses, or projection operators. In network forecasting, enforcing such constraints is vital, as system targets are strictly bounded by operational capacities. Unfeasible predictions, regardless of aggregate error metrics, can trigger catastrophic downstream decisions.

Contemporary constraint-aware frameworks predominantly rely on physics-guided loss regularization [24], [25] or reactive post-hoc clipping [26], [27]. However, soft penalties do not guarantee compliance, and post-hoc clipping creates a structural disconnect between the trained model and inference behavior. Aligning with the philosophy of structural physics induction [28], the PI-DCH layer internalizes feasibility operations directly within the forecasting head. By utilizing smooth barrier functions to shape predictions toward a dynamic capacity envelope prior to a final non-linear projection, the constraint pathway becomes an explicit, differentiable component of the computational graph.

F. Position of This Work

PhyST-Net synthesizes advancements in Transformer forecasting, adaptive graph learning, and interpretable exogenous conditioning. Its distinguishing factor lies in the rigorous organization of the forecasting interface. Rather than presenting a bespoke, single-domain architecture, this paper formulates a generalized ExG-aware graph prediction framework, subsequently instantiating physically meaningful, domain-specific modules. This methodology positions PhysST-Net as a reusable framework for domain-constrained network forecasting.

III. Problem Definition

We formulate the task as exogenous-variable-aware (ExG-aware) spatio-temporal graph forecasting. This formulation aims to predict future states of a networked system by integrating historical observations, graph-structured dependencies, and external driving factors.

A. Graph Representation and Spatial Dependencies

Let $G = (V, E, A)$ denote a spatio-temporal graph, where $V = \{v_1, \dots, v_N\}$ is the set of N nodes representing physical or logical entities (e.g., wind turbines in a farm or sensors in a city). E is the set of edges representing connections, and $A \in \mathbb{R}^{N\times N}$ is the adjacency matrix encoding spatial relations. In practice, A can be a physical adjacency matrix, a distance-based kernel, or a learned correlation matrix. To maintain flexibility across different domains, we further define M_G as a set of graph metadata, containing auxiliary information such as node coordinates, static attributes, or environmental descriptors that are not captured by the topology A alone.

B. Temporal Tensors and Exogenous Variables

At a given forecasting origin t, we define the historical target tensor $\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T\times N\times D_x}$, where T is the look-back window and D_x is the dimension of the target state at each node. Our goal is to predict the future target tensor $\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H\times N\times D_y}$ over a forecast horizon H.

A critical challenge in modern forecasting is the integration of exogenous variables (ExG), which we categorize into two distinct types based on their temporal availability:

- **Historical ExG** ($\mathbf{E}^h_{t-T+1:t} \in \mathbb{R}^{T\times D_h}$): These variables represent external conditions (e.g., historical ambient temperature or humidity) that were observed alongside the target state history. They provide contextual information to help the model identify the latent state of the system during the look-back period.
- **Future ExG** ($\mathbf{E}^f_{t+1:t+H} \in \mathbb{R}^{H\times D_f}$): These represent known or forecastable future drivers (e.g., weather forecasts or scheduled operational constraints) that are available for the duration of the forecast horizon. Unlike historical variables, future ExG serves as a conditional driver for the unknown future trajectory.

C. Operational Forecasting Assumptions

To ensure the model is applicable to real-world operational settings, we enforce a strict non-leakage constraint. At inference time t, the model f_θ is only permitted to access information available up to t for target states and historical ExG, and only those future ExG variables that are genuinely known or forecast by external sources (e.g., Numerical Weather Prediction models). This ensures that $\mathbf{E}^f$ does not contain any "future-peeking" information that would be unavailable in a deployed environment.

D. General Optimization Objective

To simplify the notation, we encapsulate all input drivers available at time t into a single input tuple $\mathcal{X}_t$:

$$\mathcal{X}_t \triangleq \big(\mathbf{X}_{t-T+1:t}, \mathbf{E}^h_{t-T+1:t}, \mathbf{E}^f_{t+1:t+H}, A, M_G\big). \tag{1}$$

The forecasting task is defined by a parameterized model f_θ which maps $\mathcal{X}_t$ to the predicted horizon $\hat{\mathbf{Y}}_{t+1:t+H}$. For domains where physical laws or operational limits must be satisfied, the model can be decomposed into an unconstrained spatio-temporal backbone $\mathcal{H}_\theta$ and a constraint operator $\mathcal{C}_\Omega$ parameterized by domain knowledge Ω:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{C}_\Omega\big(\mathcal{H}_\theta(\mathcal{X}_t)\big). \tag{2}$$

The operator $\mathcal{C}_\Omega$ ensures that the final output adheres to feasibility requirements such as non-negativity or capacity limits.

Given a training dataset $\mathcal{D}$, the model parameters θ are optimized by minimizing the Mean Absolute Error (MAE) between the predictions and the ground truth:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|\cdot H\cdot N} \sum_{(\mathcal{X},\mathbf{Y})\in\mathcal{D}} \sum_{h=1}^{H}\sum_{n=1}^{N} |f_\theta(\mathcal{X})_{h,n} - \mathbf{Y}_{h,n}|, \tag{3}$$

where the L_1 loss provides robustness against outliers in the spatio-temporal data. This objective provides a general training framework that can be easily extended with additional task-specific regularizers in later stages of the methodology.

IV. Method

PhyST-Net is an ExG-aware spatio-temporal graph Transformer designed for network-structured node forecasting with known external drivers and domain constraints. Given the inputs defined in Section III, the model follows the pipeline in Fig. 1. Historical target states and historical ExG are first embedded into node-level temporal representations. AGSP converts the temporal history into continuous tokens. A Transformer backbone learns temporal dependencies. R-STMF serves as a graph adapter that maps temporal node states to graph-enhanced node states. K-AMC injects future ExG into the latent representation. PI-DCH finally decodes the conditioned states into feasible forecasts respecting domain-specific bounds.

A. Overall Framework

For a mini-batch, let $\mathbf{X} \in \mathbb{R}^{B\times T\times N\times D_x}$ denote historical target states, $\mathbf{E}^h \in \mathbb{R}^{B\times T\times D_h}$ denote historical ExG, and $\mathbf{E}^f \in \mathbb{R}^{B\times H\times D_f}$ denote future ExG. The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{B\times H\times N\times D_y}$. The computation can be written as

$$Z = \Phi_{patch}(\mathbf{X}, \mathbf{E}^h, M_G), \tag{4}$$

$$H_T = \Phi_{temp}(Z), \quad H_G = \Phi_{graph}(H_T, A, M_G), \tag{5}$$

$$H_F = \Phi_{exg}(H_G, \mathbf{E}^f), \quad \hat{\mathbf{Y}} = \Phi_{head}(H_F, \mathbf{E}^f, \Omega), \tag{6}$$

where Φ_{patch} is AGSP, Φ_{temp} is the temporal Transformer encoder, Φ_{graph} is the graph adapter (R-STMF), Φ_{exg} is

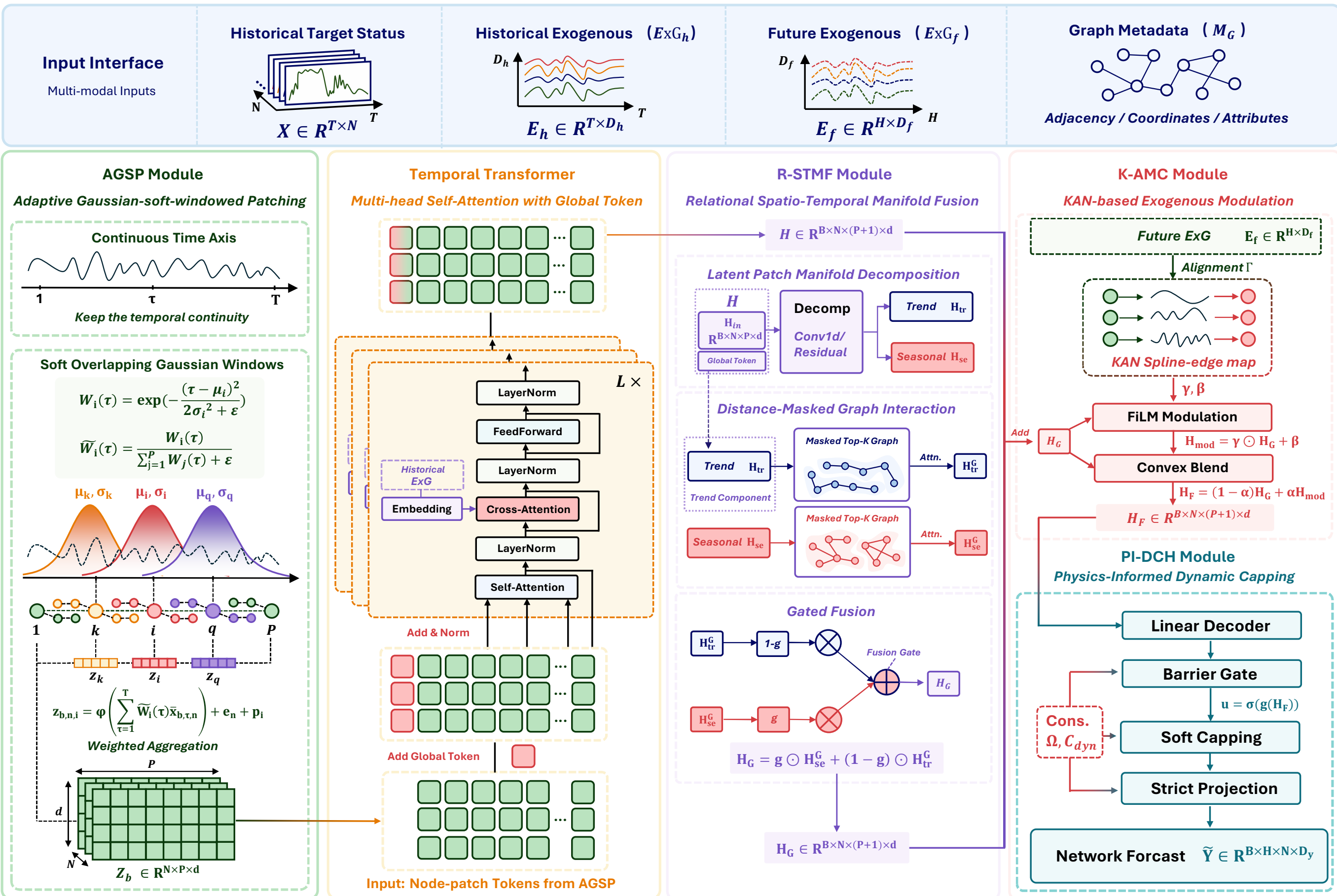

Fig. 1. Overall architecture of PhyST-Net. The framework operates as an exogenous-aware vertical pipeline: (I) Multi-modal input interface; (II) AGSP continuous tokenization; (III) Transformer sequence encoding; (IV) R-STMF relational manifold fusion; (V) K-AMC spline-based conditioning; and (VI) PI-DCH domain-constrained forecasting. All external data (ExG and Metadata) and technical details are organized in peripheral tracks to ensure a clear and interpretable data flow.

future-ExG conditioning (K-AMC), and Φ_{head} is the physics-informed prediction head (PI-DCH). This decomposition emphasizes that the architectural backbone is defined on generic graph-structured sequences, while domain-specific knowledge is encapsulated in the input interface and the constraint head.

B. Adaptive Gaussian-soft-windowed Patching

Transformer-based forecasters conventionally mitigate sequence length by aggregating observations into discrete temporal patches. While computationally convenient, fixed non-overlapping patches inevitably induce boundary artifacts when the underlying physical signal transverses a patch boundary. Aligning with emerging principles in complexity-balanced sequence modeling—where representational capacity is dynamically distributed proportional to signal variation—AGSP supersedes rigid patch assignments with learnable Gaussian soft-windows. This differentiable formulation empowers the architecture to autonomously allocate dense receptive fields to highly transient events and broader, smoother fields to stable operational regimes.

For node n and temporal window i, the unnormalized

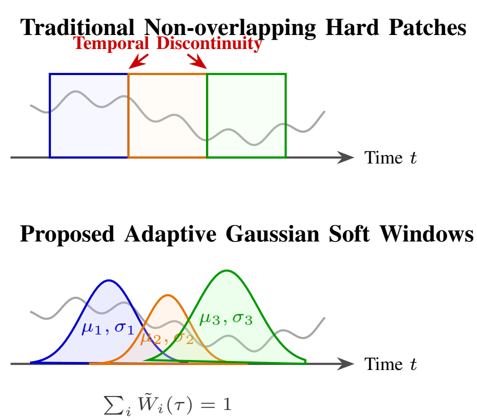

Fig. 2. Mechanics of AGSP. By replacing discrete sequence chunking with continuous, differentiable density windows, the model autonomously mitigates boundary artifacts and preserves temporal continuity. Each soft window is parameterized by a learnable center μ_i and bandwidth σ_i, with overlapping weights dynamically normalized.

weight at time index $\tau \in \{1, \ldots, T\}$ is

$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right), \tag{7}$$

where μ_i and σ_i are learnable center and bandwidth parameters. To ensure numerical stability and valid window properties, we parameterize the bandwidth as $\sigma_i = \text{softplus}(\rho_i) + \delta$, where ρ_i is the learnable parameter. The centers μ_i are initialized to uniform grid locations across the historical window to encourage an ordered temporal partition. The weights are normalized across P windows:

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^{P} W_j(\tau) + \epsilon}. \tag{8}$$

Let $\bar{x}_n(\tau)$ denote the embedded historical state. The token for node n and patch i is

$$z_{n,i} = \psi\left(\sum_{\tau=1}^{T} \tilde{W}_i(\tau)\bar{x}_n(\tau)\right) + e_n + p_i, \tag{9}$$

where $\psi(\cdot)$ is a linear projection, e_n is a node embedding, and p_i is a temporal patch embedding.

C. Temporal Representation Learning

Adopting the global token philosophy [14] to facilitate efficient long-range context fusion, the token tensor Z is concatenated with a learnable global token $z_{global} \in \mathbb{R}^{B\times N\times 1\times d}$ for each node. This global token serves as a sequence-level memory bottleneck, aggregating the macroscopic temporal representation (e.g., shared external regimes or system-wide traffic trends) across all local patches.

We denote the temporal encoding backbone as $\text{STEnc}(\cdot)$. In PhyST-Net, rather than a naive standard Transformer, STEnc incorporates a targeted cross-attention mechanism. Specifically, within each encoder layer, the augmented node-wise token sequence $[Z \parallel z_{global}]$ first undergoes standard multi-head self-attention. Subsequently, to enrich the macro-trend representation with system-wide context, the global token uniquely cross-attends to the embedded historical exogenous features (e.g., SCADA and time-mark metadata), acting as an information sink:

$$z'_{global} = z_{global} + \text{CrossAttention}(z_{global}, \mathbf{E}^h, \mathbf{E}^h). \tag{10}$$

This contextually enriched sequence is then processed through the feed-forward networks, yielding the final temporal representation:

$$H_T = \text{STEnc}([Z \parallel z_{global}]), \quad H_T \in \mathbb{R}^{B\times N\times(P+1)\times d}. \tag{11}$$

Crucially, rather than discarding the global token post-encoding, PhyST-Net structurally leverages it to separate the node's state into two distinct manifolds: a local-variation patch state $U \triangleq H_{T,1:P}$ and a macro-trend global state $R \triangleq H_{T,P+1}$.

D. Relational Spatio-Temporal Manifold Fusion (R-STMF)

R-STMF serves as the graph adapter, specifically designed to capture the multi-modal spatial dependencies inherent in physical networks. Unlike conventional graph operators that treat node features as monolithic vectors, R-STMF recognizes that spatial interactions occur across different temporal scales within the latent space. To address this, we perform a **Latent Patch Manifold Decomposition**.

Let $H_{in} = H_T$ denote the input latent representation. For each node n, we apply a 1D depth-wise convolution along the patch dimension to extract the smoothed trend component H_{tr}, while the residual represents the seasonal (oscillatory) component H_{se}:

$$H_{tr} = \text{GELU}(\text{Conv1d}(H_{in})), \tag{12}$$
$$H_{se} = H_{in} - H_{tr}. \tag{13}$$

This decomposition ensures that the model can independently learn how macro-scale trends and micro-scale seasonal fluctuations propagate through the graph.

1) Distance-Masked Relational Pruning: For each manifold $c \in \{se, tr\}$, we compute relational attention scores across nodes. To enforce physical realism and reduce computational overhead, we introduce a **Distance-Masked Top-K Pruning** strategy. Given a physical distance matrix or prior topology, we define a binary mask $\mathcal{M}_{dist}$, where $\mathcal{M}_{ij} = 1$ only if node j is within a feasible interaction radius of node i. The masked attention score S_c is:

$$S_{c,ij} = \begin{cases} \frac{Q_c(K_c)^\top}{\sqrt{d_m}} & \text{if } \mathcal{M}_{ij} = 1 \\ -\infty & \text{otherwise} \end{cases}, \quad A_c = \text{Softmax}\left(\text{TopK}(S_c, K_c)\right). \tag{14}$$

Subsequently, we perform Top-K selection on the remaining scores to ensure a sparse and robust graph routing. Finally, the graph-enhanced seasonal and trend manifolds are recombined using a learnable gated fusion mechanism:

$$g = \sigma(\text{Linear}([H^G_{se} \parallel H^G_{tr}])), \tag{15}$$
$$H_G = g \odot H^G_{se} + (1-g) \odot H^G_{tr}, \tag{16}$$

where H^G_c denotes the manifold c after spatial propagation via A_c. This dual-branch architecture allows PhyST-Net to model complex spatio-temporal interactions without conflating multi-resolution physical mechanisms.

E. Future-ExG Conditioning (K-AMC)

We refer to this module as K-AMC, short for KAN-enhanced Adaptive Modulation and Conditioning. It serves as a generalized interface for known systemic drivers, injecting future exogenous drivers $\mathbf{E}^f \in \mathbb{R}^{B\times H\times D_f}$ into the latent representations. Aligning with the TimeXer philosophy [14], the macro-trend global token acts as the primary bridge to cross-attend and ingest external factors. We define an alignment operator Γ that maps the future horizon into variate tokens corresponding to the latent states.

The aligned representation $\bar{\mathbf{E}}^f$ is then passed through a Kolmogorov-Arnold Network (KAN) block. Unlike standard MLPs, KANs parameterize transformations using learnable univariate B-spline functions on the edges. For an input feature x, the edge function is defined as:

$$\phi(x) = w_b\,\text{SiLU}(x) + \sum_{i=1}^{k} c_i B_i(x), \tag{17}$$

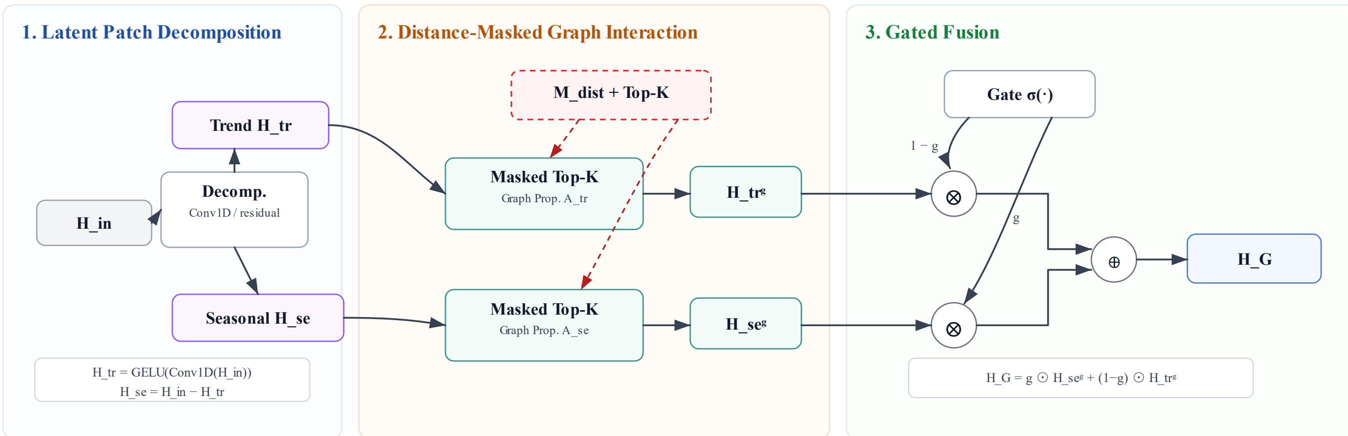

Fig. 3. Detailed architecture of R-STMF. The latent representation is decomposed into trend and seasonal manifolds. Each branch performs distance-masked Top-K graph propagation with branch-specific routing, and the resulting graph-enhanced manifolds are fused through a learnable gate.

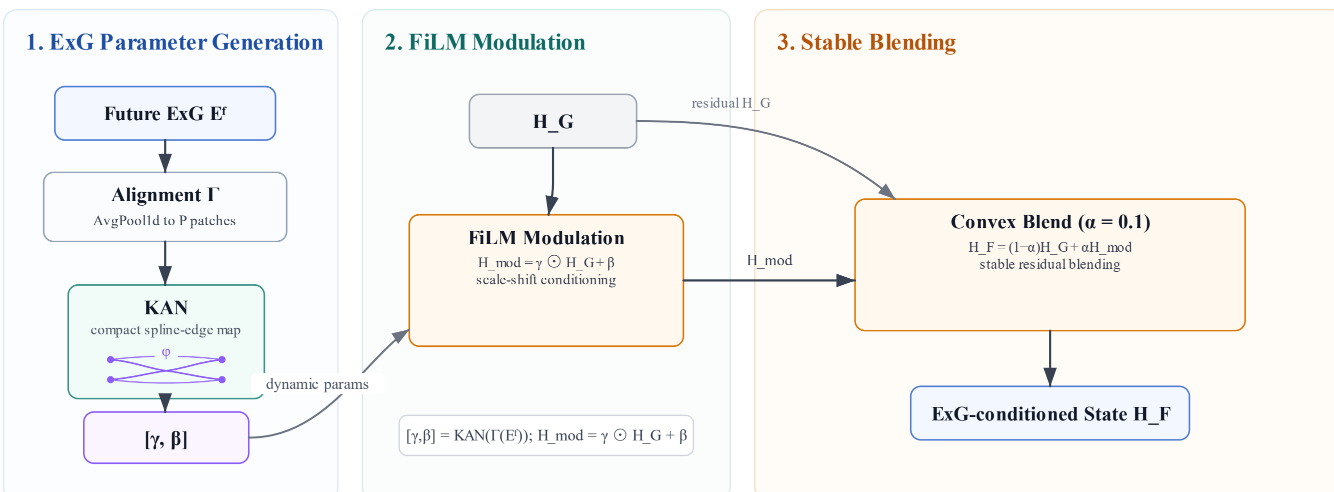

Fig. 4. Mechanics of K-AMC. Future exogenous drivers are aligned to the patch scale and processed by a compact KAN block to generate dynamic modulation parameters. The latent graph state is modulated by FiLM and stabilized through convex blending with the original state.

where $B_i(x)$ are B-spline basis functions. The analytical derivative is:

$$\frac{\partial \phi(x)}{\partial x} = w_b \, \mathrm{SiLU}'(x) + \sum_{i=1}^{k} c_i B_i'(x). \tag{18}$$

By aggregating these edge functions, the KAN block produces modulation parameters $[\gamma, \beta] = \mathrm{KAN}(\bar{\mathbf{E}}^f)$. The final conditioned latent state is then obtained via element-wise scale and shift, stabilized by a convex blending scheme with hyperparameter $\alpha = 0.1$:

$$H_{mod} = \gamma \odot H_G + \beta, \quad H_F = (1-\alpha) H_G + \alpha H_{mod}. \tag{19}$$

F. Physics-Informed Dynamic Capping Head (PI-DCH)

PI-DCH instantiates the constraint operator $\mathcal{C}_\Omega$. In mission-critical environments such as smart grids, safety-critical traffic routing, and industrial control systems, deploying predictive models mandates rigorous output-space guidance to preclude physically unfeasible trajectories. While hard projection operators (e.g., clipping) guarantee bounded outputs, they may create a mismatch between training and inference, and can severely weaken gradient signals when used naively, hampering the learning process.

To ensure that the model remains physically consistent even under extreme scenarios, we adopt a dual-stage enforcement strategy. While $\hat{Y}_{soft} = \tilde{Y} + u \odot (C_{dyn} - \tilde{Y})$ provides the differentiable path necessary for optimization via a learned barrier gate $u = \sigma(g(H_F))$, we define the final deployment-ready forecast $\hat{\mathbf{Y}}$ through a **Strict Operational Boundary Enforcement** layer:

$$\hat{\mathbf{Y}} = \mathrm{Proj}_{\mathcal{K}}(\hat{Y}_{soft}) \triangleq \max(0, \min(\hat{Y}_{soft}, C_{dyn})), \tag{20}$$

where $\mathrm{Proj}_{\mathcal{K}}$ denotes the Euclidean projection onto the dynamic feasible set $\mathcal{K} = [0, C_{dyn}]$. Crucially, this projection is treated as a non-trainable safety layer during inference to guarantee 100% compliance with physical limits. By decoupling the differentiable steering ($\hat{Y}_{soft}$) from the strict enforcement ($\hat{\mathbf{Y}}$), PhyST-Net achieves the dual goals of stable gradient backpropagation and absolute operational feasibility.

G. Computational Complexity Analysis

A core motivation of PhyST-Net is mitigating the severe computational bottlenecks of dense spatio-temporal Transformers, which conventionally exhibit a time complexity of $\mathcal{O}(N \cdot T^2 + T \cdot N^2)$ for sequence length T and node count N. PhyST-Net structurally resolves this through its decoupled topology.

In the temporal pathway, the AGSP module compresses the raw T-length sequence into P continuous tokens (where $P \ll T$). Consequently, the temporal Transformer encoder's self-attention complexity is reduced from $\mathcal{O}(N \cdot T^2)$ to $\mathcal{O}(N \cdot P^2)$.

In the spatial pathway, standard full-graph attention requires $\mathcal{O}(P \cdot N^2)$ operations. The R-STMF module introduces a Top-K sparse masking strategy (Eq. 14), restricting message passing to a parameterized neighborhood size K. This bounds the spatial complexity to $\mathcal{O}(P \cdot K \cdot N)$. Thus, the overall self-attention complexity of the PhyST-Net backbone is decoupled and bounded by $\mathcal{O}(N \cdot P^2 + P \cdot K \cdot N)$, rendering the architecture highly scalable for large physical networks and extended forecasting horizons without sacrificing expressive capability.

H. Optimization Objective

To bridge the gap between purely data-driven forecasting and physical feasibility, we minimize a multi-objective loss $\mathcal{L} = \mathcal{L}_{mae} + \lambda_\Omega \mathcal{L}_{penalty}$. The penalty term $\mathcal{L}_{penalty}$ structurally regularizes the model to internalize the operational set Ω during backpropagation. Utilizing the positive-part operator $[x]^+ \triangleq \max(0, x)$, we decouple the penalty into two physically meaningful components: a non-negativity constraint and a dynamic capacity limit constraint:

$$\mathcal{L}_{penalty} = \mathcal{L}_{neg} + \mathcal{L}_{cap}, \tag{21}$$

$$\mathcal{L}_{neg} = \frac{1}{|\mathcal{D}|HN} \sum_{\mathcal{D}} \sum_{n,h} [-\hat{Y}_{soft,n,h}]^+, \tag{22}$$

$$\mathcal{L}_{cap} = \frac{1}{|\mathcal{D}|HN} \sum_{\mathcal{D}} \sum_{n,h} [\hat{Y}_{soft,n,h} - C_{dyn,n,h}]^+. \tag{23}$$

This decoupled formulation ensures the model learns to respect physical boundaries natively, thereby reducing the reliance on reactive clipping at inference time. The overall training logic is summarized in Algorithm 1.

Algorithm 1 PhyST-Net Forward Pass

Require: Input tuple $\mathcal{X}_t = (\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G)$, Domain knowledge Ω.
Ensure: Feasible network forecast $\hat{\mathbf{Y}}$.
{*Phase I: AGSP Continuous Tokenization*}
Initialize learnable Gaussian parameters μ_i, σ_i for P tokens using metadata M_G.
for each token $i \in \{1, \dots, P\}$ **do**
  Compute soft-window weights $W_i(\tau) = \exp\left(-\frac{(\tau-\mu_i)^2}{2\sigma_i^2+\epsilon}\right)$.
  Aggregate $\mathbf{X}$ and $\mathbf{E}^h$ via $W_i(\tau)$ to form continuous token $Z_{:,i,:}$.
end for
{*Phase II: Temporal Encoding ($\mathcal{H}_\theta$ part 1)*}
Process Z via Multi-Head Self-Attention to capture temporal dependencies $\rightarrow H_T$.
{*Phase III: R-STMF Graph Adaptation ($\mathcal{H}_\theta$ part 2)*}
Decompose $H_{in} = H_T$ into seasonal H_{se} and trend H_{tr} manifolds.
Apply Top-K sparse masking to A based on spatial similarities and physical distances.
Propagate H_{se} and H_{tr} over the masked graph and fuse $\rightarrow H_G$.
{*Phase IV: K-AMC Exogenous Modulation ($\mathcal{H}_\theta$ part 3)*}
Align horizon-wise $\mathbf{E}^f$ to patch-wise representation $\bar{\mathbf{E}}^f$.
Compute B-spline edge functions via KAN block to generate $[\gamma, \beta]$.
Modulate latent states: $H_F \leftarrow (1-\alpha)H_G + \alpha(\gamma \odot H_G + \beta)$.
{*Phase V: PI-DCH Constraint Decoding ($\mathcal{C}_\Omega$)*}
Decode unconstrained prediction $\tilde{Y} = \text{Linear}(H_F)$.
Shape prediction softly: $\hat{Y}_{soft} \leftarrow \tilde{Y} + u \odot (C_{dyn} - \tilde{Y})$.
Project to feasible set: $\hat{\mathbf{Y}} \leftarrow \text{Proj}_{\mathcal{K}}(\hat{Y}_{soft})$.
return $\hat{\mathbf{Y}}$.

V. Experiments

This section evaluates PhyST-Net on wind-power forecasting tasks where both predictive accuracy and operational feasibility are important. Following the formulation in Section III, each sample contains historical turbine-level target states, available historical covariates, forecast-horizon exogenous drivers, and graph metadata. The target is active power over a future horizon. Unless otherwise stated, all reported errors are computed on the held-out test split after inverse transformation to physical units.

A. Experimental Protocol

We evaluate two operational horizons: short-term and medium-term forecasting. These settings are selected because the physical spatio-temporal interface is expected to be most informative when exogenous forcing, graph propagation, and constraint-aware decoding jointly affect the forecast. The main comparison therefore focuses on short- and medium-horizon tasks on three wind-power datasets with different spatial scales: SDWPF, HoT, and Kelmarsh.

SDWPF is a Chinese wind-farm dataset from Shandong with 134 turbines sampled every 15 minutes over more than two years. The loader supports both the historical HDF5 files and the normalized CSV layout, including turbine-level power targets, SCADA measurements, numerical weather prediction (NWP) variables, and turbine positions. The current SDWPF configuration uses two primary meteorological features, such as average wind speed and average wind direction, together with standardized coordinates for spatial encoding. HoT is the Hill of Towie wind-farm dataset from Scotland, containing 21 turbines, 192794 timestamps, five exogenous variables, and 10-minute records from 2020-01-01 to 2024-09-01; the target missing rate is 1.83%. Kelmarsh is a UK wind-farm

TABLE II
EXPERIMENTAL DESIGN FOR REPRESENTATIVE WIND-POWER TASKS.

Dataset	N	D_f	**Freq.**	**Horizon**	T	H
HoT	21	5	10 min	Short	576	432
HoT	21	5	10 min	Medium	1920	1440
Kelmarsh	6	12	10 min	Short	576	432
Kelmarsh	6	12	10 min	Medium	1920	1440
SDWPF	134	2	15 min	Short	384	288
SDWPF	134	2	15 min	Medium	1280	960

dataset with six turbines, 157536 timestamps, 12 exogenous variables, and 10-minute records from 2016-01-03 to 2018-12-31; the target matrix contains no missing values in the current processed version. All three datasets follow a common wind-task interface with target, SCADA, NWP, and position files.

All datasets are split chronologically to avoid temporal leakage. The first 70% of samples are used for training, the following 10% for validation and early stopping, and the final 20% for testing. Missing target values are filled before scaling, using forward filling followed by zero filling for residual missing entries, while original missingness can be retained by the data loader as a mask when available. Standardization parameters for power targets are fitted on the training segment and reused for validation and testing. Meteorological variables are scaled consistently across historical SCADA and forecast-horizon NWP inputs, and turbine coordinates are standardized before entering the spatial module. Calendar encodings include month, day, weekday, hour, and the native intra-hour bucket for the corresponding sampling interval.

The current repository also contains loaders and exploratory summaries for broader forecasting tasks, including GEFCom2014 Solar, GEFCom2014 Load, HK Rooftop PV, and ASHRAE building load. These datasets cover solar generation, load forecasting, rooftop PV, and building energy demand, providing additional cross-domain forecasting contexts beyond the three wind-power benchmarks emphasized in this section.

Table II summarizes the representative configurations used in the main comparison. HoT and Kelmarsh are sampled every 10 minutes, while SDWPF is sampled every 15 minutes. For 10-minute datasets, the short-term setting uses a 4-day look-back window and a 3-day forecast horizon; the medium-term setting uses about 13.3 days of history and 10 days of prediction. For SDWPF, the corresponding step counts are adjusted to its 15-minute sampling interval. This design keeps the physical duration comparable across datasets while respecting their native resolutions.

PhyST-Net uses a 3-layer encoder, one decoder layer, hidden size 256, feed-forward size 512, eight attention heads, dropout 0.1, and Adam optimization. Training is capped at 50 epochs with early stopping patience 5 and gradient clipping at 1.0. The loss combines mean-squared prediction error with a high-output penalty weighted by 2.0 to emphasize rated-power operating regimes. For the full model, AGSP, R-STMF, and K-AMC are enabled throughout the main comparison. The ablation scripts keep the same data split, optimizer, horizon, and model width, changing only the target module switch.

We compare PhyST-Net with strong recent time-series and spatio-temporal baselines, including TimeXer, CrossFormer, PatchTST, TiDE, iTransformer, Amplifier, DAG, GCGNet, DUET, TimeKAN, and CrossLinear. Models that do not natively consume exogenous wind variables are evaluated through the same historical-power and future-NWP protocol with a lightweight convolutional exogenous adapter, so that the comparison uses the same available information. The primary metrics are mean absolute error (MAE) and root mean square error (RMSE). Lower values indicate better predictive accuracy.

B. Main Results

Table III summarizes the representative benchmark results. PhyST-Net obtains the best MAE in all selected settings and the best RMSE in four out of six settings. The gains are especially pronounced on Kelmarsh, where PhyST-Net reduces MAE by 30.99% and 25.05% over the strongest competing baseline in the medium- and short-term tasks, respectively. On SDWPF, the advantage remains consistent, with MAE reductions of 21.44% and 14.06% for the medium- and short-term tasks. On HoT, PhyST-Net provides the lowest MAE in both short- and medium-term forecasting, while DAG and TimeKAN remain close competitors in RMSE. These results suggest that the proposed decomposition is most effective when the forecast must integrate non-local spatial effects and future exogenous conditions rather than relying only on near-term autoregressive continuity.

C. Ablation Study on SDWPF Short-Term Forecasting

We conduct the ablation study on SDWPF short-term forecasting because it provides a representative diagnostic setting with rich spatial relations and an appropriate prediction length. The 134-turbine network exposes non-trivial cross-node dependencies, while the 3-day forecast horizon is long enough for spatial propagation and future meteorological drivers to affect the forecast, but not so long that the evaluation is dominated only by accumulated uncertainty. This setting therefore offers a focused stress test for adaptive patching, graph fusion, and exogenous conditioning.

Table IV reports the full model and four controlled variants using the current ablation logs. The anchor model is the full PhyST-Net configuration. The no-DCP variant removes the former dynamic continuous patching branch, corresponding to the ablation of the adaptive patching component. The no-fusion variant disables the latent fusion path. The no-PSTG variant removes the former spatio-temporal graph adapter, corresponding to the R-STMF pathway. The no-K-FiLM variant removes the former exogenous modulation branch, corresponding to the K-AMC conditioning pathway. We report MAE, RMSE, R^2, and the grid-assessment Lambda accuracy used by the operational evaluation script.

The anchor model obtains the lowest MAE and RMSE, with a Lambda accuracy of 90.8948%, exceeding the 83.0% short-term assessment threshold. Removing the adaptive patching branch increases RMSE by 1.25% and slightly reduces Lambda accuracy. Removing only the fusion path produces a smaller RMSE degradation, suggesting that the backbone

TABLE III

REPRESENTATIVE FORECASTING RESULTS ON STRONG SETTINGS. BEST RESULTS ARE IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED.

Dataset	Horizon	PhyST-Net		TimeXer		CrossFormer		PatchTST		iTransformer		Amplifier		DAG		DUET		TimeKAN	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HoT	Medium	**267.32**	**468.23**	377.48	556.27	323.40	480.46	372.19	529.96	428.58	647.79	394.77	537.81	374.66	530.49	359.42	508.00	325.33	477.37
HoT	Short	**258.55**	449.75	370.98	536.43	363.96	526.93	369.48	504.82	312.79	496.73	396.91	549.22	276.90	**445.89**	363.62	509.49	343.95	501.63
Kelmarsh	Medium	**144.55**	**208.41**	231.16	328.12	280.51	410.72	349.92	477.61	209.45	283.89	382.84	527.38	306.12	410.92	264.88	361.46	337.94	445.28
Kelmarsh	Short	**138.67**	**200.07**	202.08	285.36	277.38	397.51	299.90	404.94	186.61	273.30	348.30	475.12	185.01	276.33	290.11	377.49	329.03	433.20
SDWPF	Medium	**129.85**	**215.72**	198.01	281.86	185.61	256.27	249.13	325.73	165.29	233.61	246.56	350.77	167.78	246.62	195.13	277.77	OOM	OOM
SDWPF	Short	**126.30**	**200.86**	164.12	244.59	223.18	300.65	223.91	321.39	165.21	237.90	266.87	376.54	146.96	230.52	217.38	293.28	216.25	288.11

Notes: OOM denotes out-of-memory failure under the same evaluation environment. The table reports representative settings in which long-range spatial interaction and future exogenous conditioning are expected to be material. Full horizon-level results are kept as supplementary evidence.

TABLE IV

ABLATION STUDY ON SDWPF SHORT-TERM FORECASTING.

Variant	MAE	RMSE	R^2	Lambda	$\triangle$RMSE
PhyST-Net (anchor)	**123.139**	**201.834**	**0.7635**	**90.8948%**	0.00%
w/o DCP	123.851	204.367	0.7576	90.7713%	+1.25%
w/o fusion	123.645	202.512	0.7619	90.8637%	+0.34%
w/o PSTG	124.188	202.898	0.7610	90.8184%	+0.53%
w/o K-FiLM	126.091	206.532	0.7524	90.5421%	+2.33%

Notes: DCP, PSTG, and K-FiLM are the historical implementation names corresponding to AGSP, R-STMF, and K-AMC, respectively. Lambda denotes the operational grid-assessment accuracy reported by the evaluation script.

preserves part of the temporal signal even without explicit fusion. Removing the spatio-temporal graph adapter increases RMSE by 0.53%, while removing the exogenous modulation branch produces the largest RMSE degradation of 2.33% and the lowest Lambda accuracy among the variants. These results indicate that the short-term SDWPF task benefits most from future-exogenous conditioning, while adaptive patching and relational spatial propagation provide additional measurable gains.

D. Discussion of the Evaluation Scope

The benchmark results show that PhyST-Net is strongest in short- and medium-horizon regimes, where future meteorological drivers and spatial propagation have sufficient time to influence the target trajectory. We therefore emphasize the operating regime aligned with the design of PhyST-Net: constrained graph forecasting under exogenous forcing.

The current ablation evidence is task-specific. It demonstrates that adaptive patching, fusion, relational graph propagation, and exogenous modulation improve SDWPF short-term forecasting, complementing the cross-dataset main comparison on SDWPF, HoT, and Kelmarsh.

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